X-Ray Light Curve Midterm Project

Maggie Ireland

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1 Neutral Hydrogen Column Density through the Galaxy

1.1 Information search

The neutral hydrogen tool from NASA (url) will be used along with the given coordinates to find the neutral hydrogen in the direction of our sources.

When increasing Dec. the NH increases as R.A. stays near the measured value. As both decrease. It appears that one area has more NH than the surrounding areas from the supernova. That area may point to a larger density of a neutral hydrogen absorbing column.

2 Observation Information

Within the Chandra archive 11 observation ID's were the subject of data retrieval. For each observation the start and stop dates in universal time, modified Julian date, and days since merger will be calculated, as well as mid time of the observation. Exposure time will also be acquired.

3 Detection Methods

All observation files were given the same task functions to detect an X-ray source. first, the ciao task dmcopy was used to filter events from 500-800 eV within a bin size of [x=3700:4200:1.0, y=3800:4300:1.0]. Next mkpsfmap was used to create a psf map for the wavdetect function. The wavdetect function

	Coordinates (RA. and Dec.)	NH value cm^-2
SN2014C	22:37:05.60,+34:24:31.9	6.93E+20
GW170817	13:09:48.08,-23:22:53.34	7.60E+20

Table 1: Neutral Hydrogen data of each given source, coordinates of R.A. and Dec., and resulting NH value from the NASA tool.

was used on all observations to attempt to detect a source by returning a region where all filtered events would be shown. Within the program ds9, sources for all observations would be verified.

3.1 X-ray Sources Detected

Only 3 of the 11 observations returned significant detections of X-ray sources.

Observation	19294	20728	20860
Start time (UT)	2017-08-26 10:33:50	2017-09-01 15:22:22	2018-08-10 18:01:12
Start time (MJD)	57991.440	57997.640	58340.750
Start time (DSM)	8.911	15.112	358.222
Stop time (UT)	2017-08-27 1:17:39	2017-09-02 4:53:25	2018-08-11 13:32:24
Stop time (MJD)	57992.053	57998.203	58341.564
Stop time (DSM)	9.525	15.675	359.035
Mid time (MJD)	57991.747	57997.922	58341.157
Exposure time (sec)	4.94066E+04	4.940659E+04	6.71647E+04
0.5-8 keV count rate (counts)	17±4.123	16±4	35±6
Best fitting photon index	0.832653 (-0.593754,0.841249)	1.58045 (-0.500791,1.24505)	1.75332 (-0.354844,0.371099)
Absorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	8.0535e-15 (0.0 - 7.069e-15)	5.623e-15 (0.0 - 5.644e-15)	7.4959e-15 (3.093e-15 - 7.649e-15)
Unbsorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	8.0728e-15 (0.0 - 6.558e-15)	5.825e-15 (0.0 - 6.039e-15)	8.1521e-15 (1.937e-15 - 8.199e-15)

Table 2: Observations 18955, 20860, and 20936 where X-ray sources were not detected. Obtained information of start and stop times in different units, mid time of observation, exposure time of observation, count rate results for the source from 0.5-8 keV, best fitting photon index with uncertainty, and absorbed and unabsorbed flux for 0.3-10 keV with uncertainty

For these sources, once wavdetect had confirmed there was a significant count rate for each observation, the xspec tool was used in order to find the best fitting photon index and corresponding absorbed and unabsorbed flux for 0.3 - 10 keV including uncertainties for each result.

3.2 X-ray Sources Not Detected

The remaining 8 observations did not have an X-ray source. Wavdetect failed to detect a source, which then prompted the use of dmlist counts for each observation's filtered image file for a 2" radius around the source. None of these sources had counts significant compared to the background region, which counts were found in a 38" region away from detected sources in the chip, which was then multiplied by the ratio of the radii from the source and the background to obtain an estimate of background counts. Next an upper limit of counts was estimated by taking 3 sigma of the given uncertainty from the background region by ds9 and divided by the found exposure time. After each source was deemed a

Observation	18955	20860	20936
Start time (UT)	2017 - 08 - 19 17:10:09	2017-12-03 1:39:54	2018-01-17 21:28:45
Start time (MJD)	57984.715	58090.069	58135.894
Start time (DSM)	2.186	107.540	153.366
Stop time (UT)	2017 - 08 - 20 00:43:25	2017-12-03 23:07:27	2018-01-18 7:04:56
Stop time (MJD)	57985.030	58090.963	58136.295
Stop time (DSM)	2.501	108.434	153.766
Mid time (MJD)	57984.872	58090.516	58136.095
Exposure time (sec)	2.463E+04	7.408E+04	3.174E+04
0.5-8 keV count rate	0±0	2±1.414	2±1.414
Background count rate estimation	255±15.94	771±27.76	326±18.1
Count rate upper limit estimation at 3 σ c.l.	0.00194	0.00112	0.00171
Absorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	5.27E-14±1.76E-14	3.05E-14±1.02E-14	4.65E-14±1.55E-14
Unabsorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	6.90E-14±2.30E-14	4.00E-14±1.33E-14	6.08E-14±2.03E-14

Table 3: Observations 18955, 20860, and 20936 where X-ray sources were not detected. Obtained information of start and stop times in different units, mid time of observation, exposure time of observation, count rate results for the source from 0.5-8 keV, background count rate estimation, estimated upper limit of counts, and absorbed and unabsorbed flux for 0.3-10 keV with uncertainty

non-detection, the upper limit estimate was taken to the pimms tool by Chandra (url). This tool was able to return the values for absorbed and unabsorbed flux of 0.3 - 10 keV in physical units of (ergs/cm**2/s). The uncertainties for the obtained flux results was found by dividing the upper limit count rate by 3.

4 Light Curves

4.1 Graphical representation

Once all observations had measures of absorbed and unabsorbed flux obtained, a graph comparing the results to the previously found time data could be created.

For the absorbed flux rise fit the normalization with uncertainty is 1.068e-14± 1.005e-14, and the slope with uncertainty is 0.259± 0.235. For the decay fit the normalization with uncertainty is 2.055e-13±6.238e-13 and the slope with uncertainty is -0.300±0.537. The fitting results for the unabsorbed flux rise normalization with uncertainty is 1.145e-14±1.199e-14 and the slope with uncertainty is 0.294±0.262. The decaying fit normalization with uncertainty is 2.961e-13±9.818e-13 and the slope with uncertainty is -0.322±0.587

Observation	20937	20938	21090
Start time (UT)	2018-01-23 7:47:35	2018-01-21 13:29:10	2018-01-17 21:28:45
Start time (MJD)	58141.324	58139.561	58243.043
Start time (DSM)	158.796	157.033	260.515
Stop time (UT)	2018-01-23 14:06:27	2018-01-21 18:27:05	2018-05-05 14:38:08
Stop time (MJD)	58141.587	58139.768	58243.609
Stop time (DSM)	159.05	157.240	261.081
Mid time (MJD)	58141.456	58139.665	58243.326
Exposure time (sec)	2.077E+04	1.585E+04	4.600E+04
0.5-8 keV count rate	1±1	0±0	4±2
Background count rate estimation	209±14.45	159±12.61	405±20.12
Count rate upper limit estimation at 3 σ c.l.	0.00208	0.00238	0.00131
Absorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	5.66E-14±1.89E-14	6.48E-14±2.16E-14	3.56E-14±1.19E-14
Unabsorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	7.41E-14±2.47E-14	8.48E-14±2.83E-14	4.67E-14±1.56E-14

Table 4: Observations 20937, 20938, and 21090 where X-ray sources were not detected. Obtained information of start and stop times in different units, mid time of observation, exposure time of observation, count rate results for the source from 0.5-8 keV, background count rate estimation, estimated upper limit of counts, and absorbed and unabsorbed flux for 0.3-10 keV with uncertainty

Observation	22157	22158
Start time (UT)	2019-03-22 6:03:14	2019-03-23 22:34:42
Start time (MJD)	58564.252	58565.982
Start time (DSM)	581.723	583.453
Stop time (UT)	2019-03-22 17:08:02	2019-03-24 7:07:56
Stop time (MJD)	58564.713	58566.297
Stop time (DSM)	582.185	583.768
Mid time (MJD)	58564.483	58566.139
Exposure time (sec)	3.818E+04	2.493E+04
0.5-8 keV count rate	0±0	3±1.730
Background count rate estimation	438±21	295±17.17
Count rate upper limit estimation at 3 σ c.l.	0.00164	0.00206
Absorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	4.48E-14±1.49E-14	5.61E-14±1.87E-14
Unabsorbed flux 0.3-10 keV (ergs/cm^2/s) 1 σ c.l.	5.87E-14±1.96E-14	7.35E-14±2.45E-14

Table 5: Observations 22157 and 22158 where X-ray sources were not detected. Obtained information of start and stop times in different units, mid time of observation, exposure time of observation, count rate results for the source from 0.5-8 keV, background count rate estimation, estimated upper limit of counts, and absorbed and unabsorbed flux for 0.3-10 keV with uncertainty

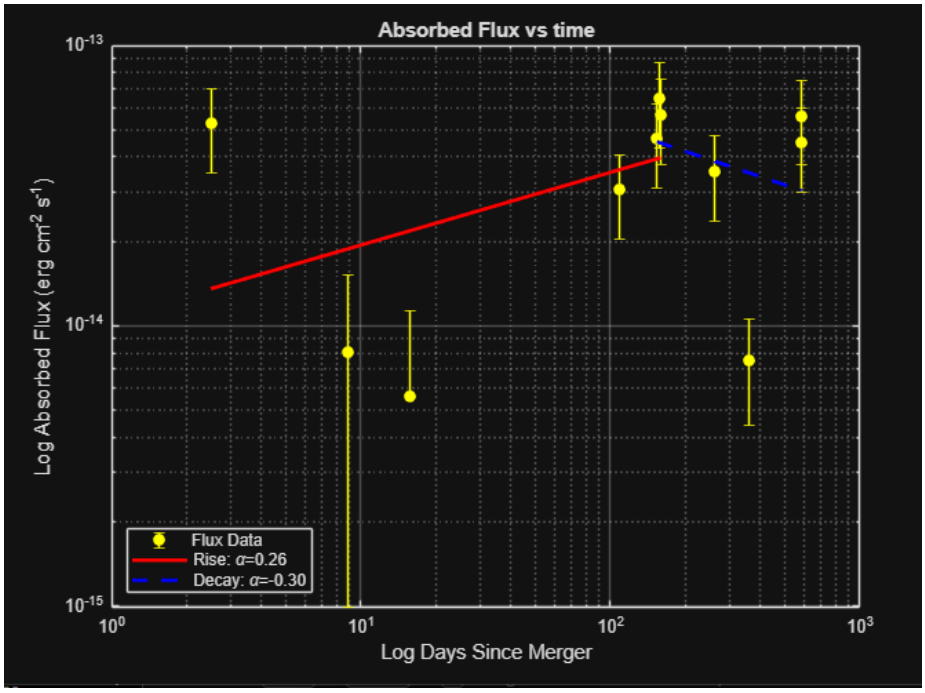


Figure 1: Resulting absorbed flux vs start of observation days since merger with two power law fits for all eleven observations. The rising normalization and slope with uncertainty are 1.068e-14± 1.005e-14 and 0.259± 0.235. The decaying normalization and slope with uncertainty are 2.055e-13±6.238e-13 and -0.300±0.537.

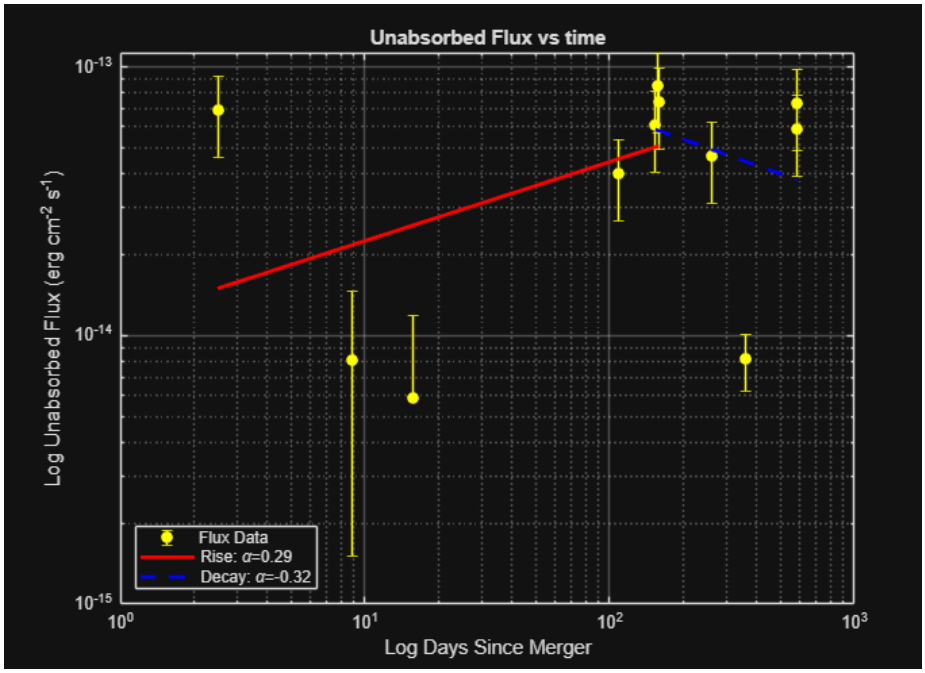


Figure 2: Resulting unabsorbed flux vs start of observation days since merger with two power law fits for all eleven observations. The rising normalization and slope with uncertainty are 1.145e-14±1.199e-14 and 0.294±0.262. The decaying normalization and slope with uncertainty are 2.961e-13±9.818e-13 and -0.322±0.587.

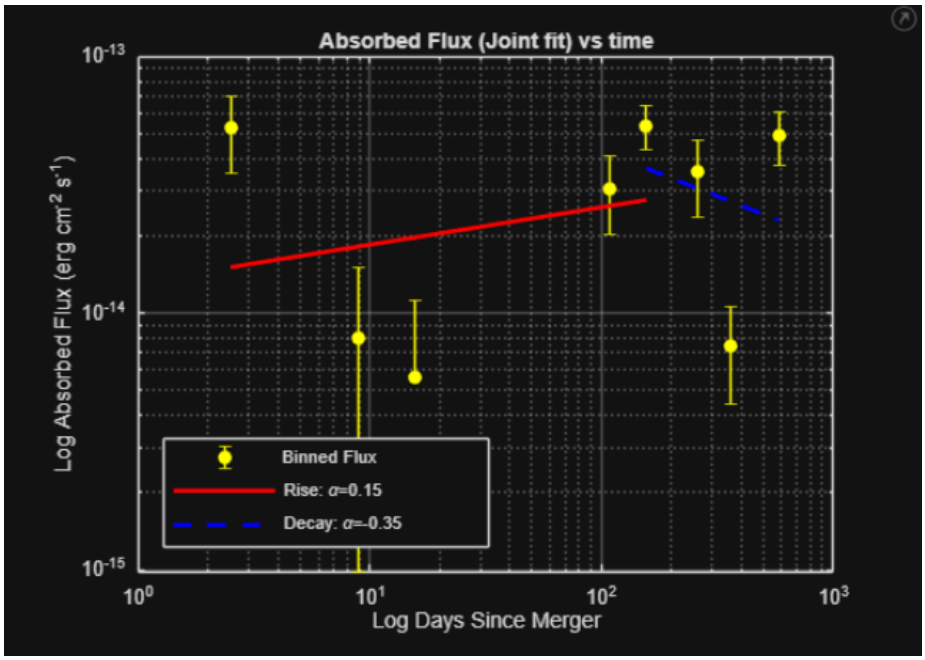


Figure 3: Resulting jointly fit absorbed flux vs start of observation days since merger with two power law fits. The rising normalization and slope with uncertainty are 1.322e-14 ± 1.583e-14 and 0.146 ± 0.344. The decaying normalization and slope with uncertainty are 2.201e-13 ± 1.450e-12 and -0.354 ± 1.148.

4.2 Further Graphical analysis

As seen in the above graphs, some points are nearly on top of each other, representing observations that are very close together in time. These can be jointly fit. Observation numbers 20936, 20937, and 20938 days since merger time only span about four days, so they need to be fit as one. Additionally, observations 22157 and 22158 are only two days apart in their days since merger measurements, so they will also be fit jointly. Doing this will better constrain the flux. New light curve plots will be made with a new fit.

For the jointly fitted absorbed flux rising fit, the normalization with uncertainty is 1.322e-14 ± 1.583e-14, while the slope with uncertainty is 0.146 ± 0.344. The decaying fit normalization with uncertainty is 2.201e-13 ± 1.450e-12, and the slope with uncertainty is -0.354 ± 1.148. For the jointly fitted unabsorbed flux rising fit, the normalization with uncertainty is 1.435e-14 ± 1.926e-14 and the slope with uncertainty is 0.174 ± 0.385. For the decaying fit, the normalization with uncertainty is 3.318e-13 ± 2.404e-12, and the slope with uncertainty is -0.387 ± 1.263.

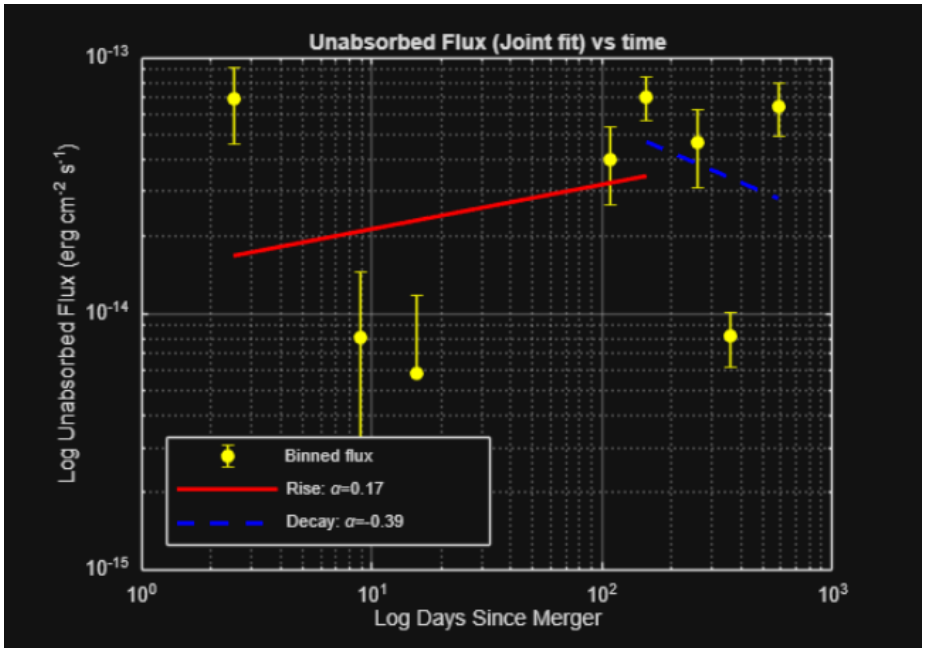


Figure 4: Resulting jointly fit unabsorbed flux vs start of observation days since merger with two power law fits. The rising normalization and slope with uncertainty are 1.435e-14 ± 1.926e-14 and 0.174 ± 0.385. The decaying normalization and slope with uncertainty are 3.318e-13 ± 2.404e-12 and -0.387 ± 1.263. .